Code No: 136EV **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
DATA STRUCTURES
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Write recursive algorithm to find factorial of a number. [2]
b) What is the space complexity of a program to insert an element into singly linked list? [3]
c) What are the applications of stack? [2]
d) Convert the given infix expression to prefix and postfix expressions: a+b/(c*d/(e-f)) [3]
e) What are properties of binary trees? [2]
f) Give an example for graph representation with adjacency matrix and adjacency list. [3]
g) How time complexity of binary search is less than linear search? [2]
h) What are the advantages and disadvantages of static hashing? [3]
i) What are splay trees? Give an example. [2]
j) What are the properties of an AVL Tree? [3]

PART – B

(50 Marks)

2. What are singly linked lists? Write an algorithm for insertion and deletion operations in singly linked lists. [10]

OR

3.a) Write an algorithm to multiply two matrices of n × m order and explain.
b) What is a recursive algorithm? Write down the recursive algorithm to generate 'n' Fibonacci series. [5+5]

4.a) Write an algorithm to evaluate a postfix expression.
b) Explain the basic operations performed on a stack. [5+5]

OR

5. What are Circular queues? Write an algorithm to perform insertion and deletion operations on circular queue. [10]

6. What are binary tree traversals? Explain with examples and also state their applications. [10]

OR

7. What is max heap? Write an algorithm to perform insertion and deletion operations from max heap. [10]

8. Explain selection sort method. [10]

OR

9. With an algorithm and example, explain binary search technique. [10]

10. What are AVL trees? Write the pseudcode to perform insert and delete operations on AVL trees and explain. [10]

OR

11. Explain insert and delete operations on B trees with example. [10]

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